Digital Payment Gateway Simulation

Class Identification, Strong / Weak Classification and Class Diagram

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University Reg. No.	202401100200098	**Assignment**	Classes & Class Diagram

Task

Identify the classes of a **Digital Payment Gateway Simulation**, classify them as **strong** and **weak** classes, and draw the class diagram.

1. Problem Overview

A digital payment gateway sits between a customer, a merchant and the banks. A customer chooses a payment method (card or UPI), the gateway creates a transaction, verifies the payment (for cards via an OTP), asks the bank to debit the customer and credit the merchant, and finally issues a receipt. If something goes wrong the merchant can raise a refund, and every step is recorded in a log for auditing.

2. Identification of Classes (Noun Analysis)

Nouns were picked from the description above. Nouns that describe a property of something else became attributes (e.g. amount, card number, expiry date, status), and vague nouns (e.g. system, payment) were dropped or merged into a real class. The classes that remain are:

#	Class	Responsibility in the simulation
1	**Customer**	Person who registers, logs in and initiates a payment.
2	**Merchant**	Business that receives the payment, gets settlement and can request refunds.
3	**PaymentGateway**	Central controller: receives the request, processes it and routes it to the bank.
4	**Bank**	Verifies available funds and performs the debit / credit.
5	**PaymentMethod** (abstract)	General idea of a way to pay; defines validate() and authorize().
6	**CardPayment**	Payment by debit / credit card; validates card details and verifies the OTP.
7	**UPIPayment**	Payment through a UPI id; validates the id and sends a collect request.
8	**Transaction**	One payment attempt with amount, time and status (Initiated / Success / Failed).
9	**Receipt**	Proof of a successful transaction given to the customer.
10	**Refund**	Return of money for a transaction that was already completed.
11	**OTP**	One-time password generated to authenticate a particular transaction.
12	**TransactionLog**	Audit entries recording each event of a transaction.

3. Strong and Weak Classes

- **Strong class:** has its own unique identifier (key attribute) and can exist on its own. Its life does not depend on any other class.
- **Weak class:** cannot be identified by its own attributes alone. It has only a partial key and is identified together with the key of its owner (a strong class). It is created with the owner, is meaningless without it and is deleted when the owner is deleted (composition).

3.1 Strong classes

Class	Identifier	Why it is strong
Customer	customerId	Registered independently; exists before any payment is made.
Merchant	merchantId	Registered independently with the gateway.
PaymentGateway	gatewayId	The core system; exists whether or not any transaction is running.
Bank	bankId	External institution that exists independently of the gateway.
PaymentMethod (CardPayment, UPIPayment)	methodId	Belongs to a customer but has its own id and is reused across many transactions.
Transaction	txnId	Has a unique transaction id; it is the hub that owns all the weak classes.

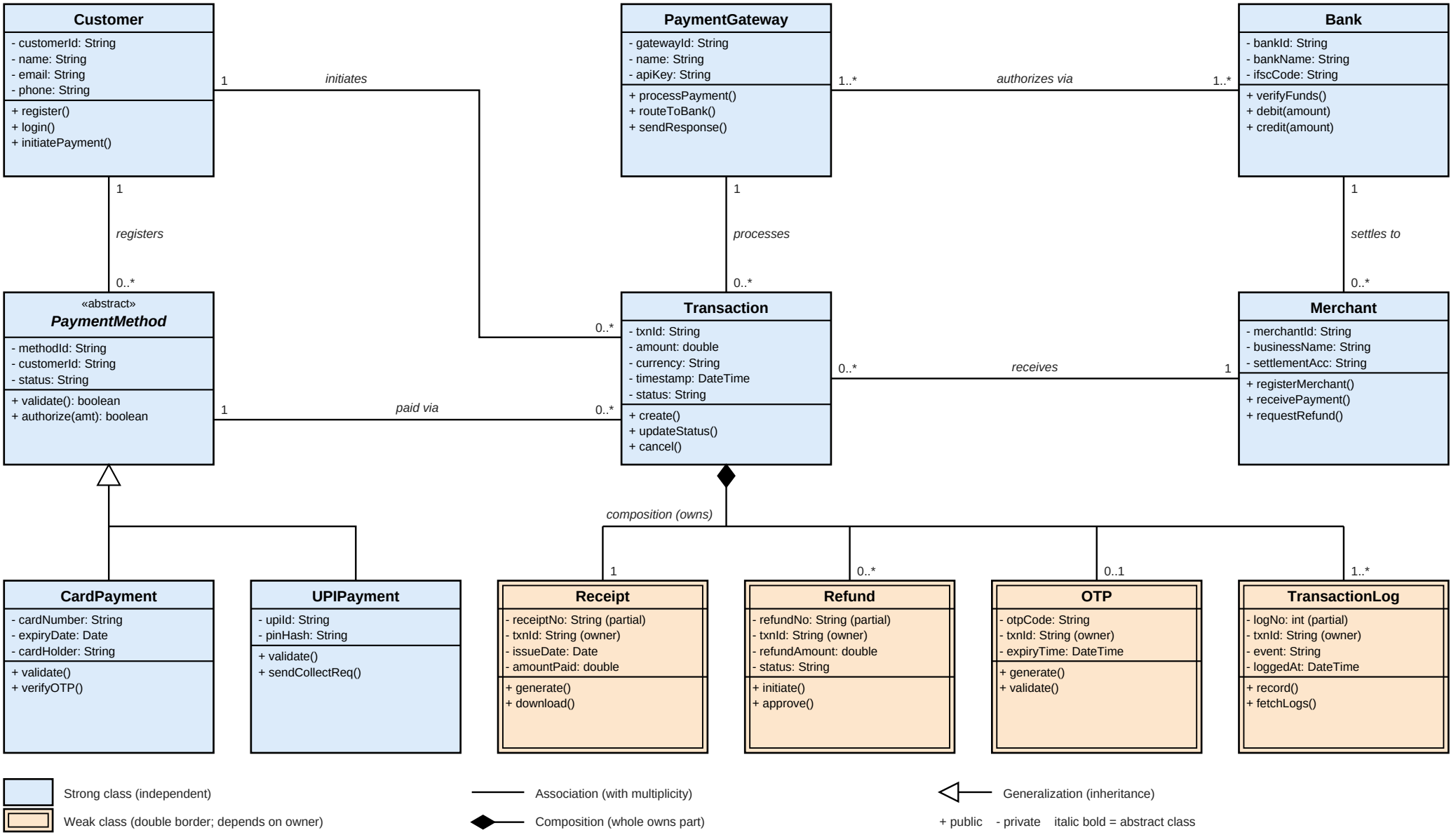
3.2 Weak classes

Class	Owner (strong)	Partial key + owner key	Why it is weak
Receipt	Transaction	receiptNo + txnId	A receipt is only meaningful for one transaction and is removed with it.
Refund	Transaction	refundNo + txnId	A refund always refers to an existing transaction; no transaction, no refund.
OTP	Transaction	otpCode + txnId	Generated for one transaction, short-lived, useless on its own.
TransactionLog	Transaction	logNo + txnId	A log line only makes sense as part of the history of its transaction.

4. Relationships Between Classes

Class A	Relationship	Class B	Multiplicity (A : B)
Customer	registers	PaymentMethod	1 : 0..*
Customer	initiates	Transaction	1 : 0..*
PaymentMethod	is used to pay (paid via)	Transaction	1 : 0..*
PaymentGateway	processes	Transaction	1 : 0..*
PaymentGateway	authorizes via	Bank	1..* : 1..*
Bank	settles to	Merchant	1 : 0..*
Merchant	receives	Transaction	1 : 0..*
Transaction	**composition** (owns)	Receipt	1 : 1
Transaction	**composition** (owns)	Refund	1 : 0..*
Transaction	**composition** (owns)	OTP	1 : 0..1
Transaction	**composition** (owns)	TransactionLog	1 : 1..*
CardPayment, UPIPayment	**generalization** (is-a)	PaymentMethod	inheritance

5. Class Diagram - Digital Payment Gateway Simulation



6. How to Read the Diagram

- Each class is a box with three parts: the class name, its attributes and its methods.
- **Blue boxes** are strong classes. **Orange boxes with a double border** are weak classes; each of them carries the owner's key (*txnId*) as part of its identity.
- A **filled diamond** on Transaction shows composition: Receipt, Refund, OTP and TransactionLog are parts of a transaction and cannot live without it.
- A **hollow triangle** on PaymentMethod shows inheritance: CardPayment and UPIPayment are kinds of PaymentMethod and implement its abstract operations in their own way.
- Numbers at the ends of the lines (1, 0..*, 1..*, 0..1) are multiplicities, e.g. one customer can initiate many transactions, but a transaction is initiated by exactly one customer.

7. Conclusion

The payment gateway simulation has **12 classes**: **8 strong classes** (Customer, Merchant, PaymentGateway, Bank, PaymentMethod, CardPayment, UPIPayment, Transaction) and **4 weak classes** (Receipt, Refund, OTP, TransactionLog). The Transaction class is the centre of the design: it links the customer, merchant, gateway and payment method, and it owns every weak class through composition.